Exploring the Dynamic Link Between ESG Public Attention and ESG Investment Returns: A VECM-VAR Analysis

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1. Introduction

Investor attention to ESG and climate change issues has been recognized as a key factor in shaping ESG demand, with empirical evidence suggesting that it can drive short-term excess returns for ESG-oriented companies (El Ouadghiri et al., 2019). However, while existing studies have examined aspects of this relationship, comprehensive analysis remains lacking, leaving a critical gap in the literature that warrants further investigation.

In this study, we construct public attention indices for four ESG-related topics—General ESG Interest, ESG Investment, Renewable Energy and Climate-Related—using Google Trends search volume index from January 2020 to January 2025. A VECM-VAR model is employed to examine the dynamic relationship between public interest in these ESG subtopics and the excess returns of high-ESG-rated firms, with a particular focus on short-term interactions and long-term equilibrium dynamics. If validated, this approach could provide valuable insights for investors, policymakers, and fund managers, enabling more effective capital allocation and risk management in sustainable finance.

2. Research Motivation and Questions

As a student majoring in quantitative finance, I have long been interested in investor sentiment and its interaction with financial markets. In my previous research, I attempted to incorporate web-based investor sentiment data into quantitative investment strategies, particularly in the Taiwanese index market. For example, I utilized data from Google Trends and applied a random forest model to develop trading strategies that assess the impact of market sentiment on index futures movements. These experiences deepened my understanding of how investor sentiment influences asset pricing and enhanced my ability to integrate sentiment analysis into quantitative trading strategies.

Last semester, while taking a Corporate Finance Special Topics course, I was introduced to numerous academic papers on Environmental, Social, and Governance (ESG) issues. Through this exposure, I realized that ESG is not merely an environmental concept; rather, it reflects the collective green preferences of the public, which are increasingly integrated with corporate value. I found ESG to be an innovative and compelling field, particularly due to this concept is strongly relevance to investor sentiment—an area I have long been passionate about. After discussions with my professor, I decided to bridge these two domains and investigate the relationship between public attention and ESG corporate excess returns.

When reviewing the existing literature, I discovered that research on public attention and ESG excess returns remains underdeveloped. There do have some great studies that focus on the short-term impact, however, to the best of my knowledge, no study has examined the long-term equilibrium relationship or reinforcing effects between variables. Consequently, my research aims to address this gap by exploring the long-run dynamics of pubic attention and ESG corporate returns, as well as the short-term adjustment mechanisms. By undertaking this study, I hope to contribute to this evolving field while simultaneously enhancing my research capabilities.

This study aims to address the following key research questions:

- Does public attention to ESG-related topics exhibit a long-term equilibrium relationship with ESG investment returns?
- In the short term, does a reinforcing effect or an overreaction phenomenon occur?
- How does the impact of public attention on ESG corporate returns vary across different ESG subtopics?

3. Literature Review

3.1 The Paradox Between Expected and Actual Returns of ESG Companies

In traditional asset pricing theory, expected returns under market equilibrium conditions should reflect investors' risk exposure and return compensation. However, in recent years, a paradox has emerged regarding the discrepancy between the expected and actual returns of Environmental, Social, and Governance (ESG) investments.

Pedersen, Fitzgibbons, and Pomorski (2021) introduced the ESG-efficient frontier framework, suggesting that investors derive nonpecuniary utility from ESG investments and are therefore willing to accept lower risk-adjusted returns. This results in equilibrium expected returns for ESG assets being lower than those of traditional assets. Furthermore, Sautner et al. (2023) found that a firm's climate risk exposure influences its risk premium: firms with high climate risk typically face higher capital costs, whereas firms with high ESG ratings are perceived as more resilient and thus attract investor preference, further lowering their expected returns.

However, since the 2010s, ESG assets have generally outperformed the broader market, exhibiting excess returns. Pastor et al. (2019) provided an equilibrium model to explanation for this phenomenon, arguing that unexpected demand shocks for ESG investments are the primary driver. They proposed that an increase in ESG demand generates excess returns through two channels. First, as investors reallocate capital to ESG assets, their market value increases directly. Second, growing consumer demand for ESG products enhances corporate earnings, further impacting firm valuations.

Pastor et al.(2022) further extended this and found that climate-related negative news—such as the introduction of stricter environmental regulations or worsening reports on global warming—is highly correlated with ESG asset returns. However, if we exclude the excess returns driven by these negative news events, ESG assets actually underperform the broader market. This suggests that fluctuations in market demand for ESG investments are closely tied to public concerns about climate risks. However, a crucial question remains: how can we effectively capture public concern? This forms the central theme of the present study.

3.2 Public Attention and ESG Excess Return

The *attention hypothesis* suggests that the level of public interest in market-related information influences investment demand and capital flows (Barber & Odean, 2008). This hypothesis has been validated in various financial markets, including oil, gold (Li et al., 2022) and even Bitcoin (Aslanidis et al., 2022).

In the ESG domain, prior studies have explored similar dynamics. El Ouadghiri (2019) was the first to investigate how public concern about climate change affects the returns of sustainable indices (DJSI US and FTSE4Good USA Index) compared to traditional market indices (S&P 500 and FTSE USA). By analyzing Google search volume indices for the keywords: *climate change* and *pollution*, the study found that increased public attention to environmental issues had a short-term positive impact on sustainable investments while negatively affecting traditional market indices. Kvam (2022) expanded this research by incorporating Twitter data and found similar results. Additionally, they discovered that during periods of high market uncertainty—as proxied by the VIX index—demand for high-ESG-rated firms further increased, which is consistent with Zhang's (2019) findings that ESG demand surged during the COVID-19 pandemic. Ballinari & Mahmoud (2022) also contributed to this field; however they focused on the predictive power of investor sentiment, finding that sentiment-driven forecasts influence capital inflows into sustainable mutual funds.

While beyond studies focus on the overall ESG market, some researchers have examined the impact of investor sentiment on individual firms. For instance, Nicolas et al. (2023) used Twitter data to analyze how ESG sentiment affects corporate reputation risk, revealing that deteriorating ESG sentiment leads to a significant decline in excess returns (-0.28%). Similarly, Dorfleitner et al. (2024) applied natural language processing (NLP) techniques to analyze ESG-related news sentiment and found that positive ESG news drives short-term stock price increases, whereas negative ESG news triggers market sell-offs.

The studies most relevant to this research are those by El Ouadghiri et al. (2019) and Kvam et al. (2022). However, their models can only capture the one-way impact of current-period climate attention on next-period sustainable asset returns and thus fail to account for dynamic equilibrium over the medium and long term or potential feedback mechanisms.

To address this limitation, this study employs **Vector Error Correction Models (VECM) and Vector Autoregression (VAR)** (Engle & Granger, 1987; Johansen, 1988) to examine both short-term fluctuations and long-term interactions between public attention and ESG investment demand. Unlike traditional models VECM explicitly accounts for the presence of a long-run equilibrium by identifying cointegration relationships between variables. This is particularly important in the ESG domain, where investor sentiment and market reactions may exhibit persistent trends rather than merely transitory effects. By incorporating an error correction term (ECT), VECM not only captures the speed at which ESG investment returns and public attention revert to equilibrium following external shocks but also provides insights into the strength and persistence of their interdependencies. Furthermore, VECM enables the separation of temporary market overreactions from structural shifts in ESG investment demand, offering a more comprehensive understanding of how investor sentiment translates into capital allocation over different time horizons. This dynamic approach overcomes the limitations of previous studies by allowing for the examination of both immediate investor reactions and gradual market adjustments, thereby providing a more holistic view of ESG-related investment behavior.

Moreover, this study categorizes public attention into four topics: ESG attention, ESG investment attention, climate change attention, and sustainable energy attention. These topics are further divided into two groups—one related more directly to investments and the other to broader environmental concerns. By considering a wider range of key terms, this study aims to address gaps in prior research and provide a more comprehensive analysis of investor sentiment.

3.3 Google Trends and ESG Investment Demand

This study uses Google Trends' relative search volume index (GSVI) as a proxy for public attention due to its real-time nature and its ability to anticipate market movements. Prior research has demonstrated that GSVI effectively captures investor attention and correlates significantly with market capital flows (Da et al., 2011) and volatilities (Preis et al., 2013), suggesting that search behavior may coincide with investment decisions rather than merely reflecting information retrieval.

Additionally, compared to traditional surveys or low-frequency data, Google Trends is more suitable for monitoring short-term fluctuations in ESG-related attention (Schmidt et al., 2019). Furthermore, compared to news sentiment analysis or social media data (e.g., Twitter), Google Trends is less susceptible to misinformation or extreme sentiment, making it a more stable indicator of investor information demand (Tetlock, 2007; Bollen, Mao, & Zeng, 2011).

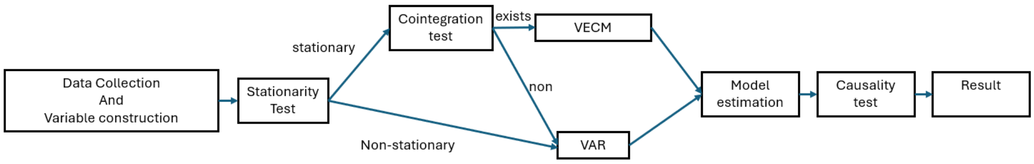
However, Google Trends has several limitations. First, its data is standardized on a 0-100 scale, meaning it reflects relative search volume rather than absolute levels of interest (Choi & Varian, 2012). Second, search trends may be influenced by Google's algorithms or external market factors, such as media coverage, policy changes, or corporate public relations strategies (Bijl et al., 2016). For example, during global climate summits there might be rise in searches for *climate change*, however the positive news generated from this kind of big event may already been absorb by the market.

To mitigate potential biases from single-word search terms, this study constructs variables using multiple key terms and incorporates external factors (e.g., VIX index) into the econometric model to enhance the accuracy of Google Trends indicators.

4. Research Procedures and Methodology

4.1 Research Procedures

This study applies standard econometric VECM procedures, including stationarity testing, cointegration analysis, and model estimation (see Figure1).



(Figure1)

4.2 Data and Variable Construction

The public interest data is sourced from Google Trends, while price data and the VIX index are obtained from Yahoo Finance. The dataset focuses on the United States and covers the period from January 2020 to January 2025.

4.2.1 ESG Search Interest Variables

To quantify public attention toward different ESG-related topics, this study constructs four distinct variables, each representing a specific ESG theme (see Table 1). For each theme, a set of relevant keywords is selected, and their corresponding weekly GSVI are retrieved. These indices are standardized by Google on a 0–100 scale, where 100 represents the peak search volume within the selected time period.

To create a more robust measure, the search interest variable for each ESG theme is computed as a weighted average of the search indices of its associated keywords. Additionally, a logarithmic transformation: $ln(GSVI + 1)$ is applied to improve the distributional properties of the data and reduce volatility.

Topic	Keywords
General ESG Interest	ESG, ESG meaning
ESG Investment	ESG/Green investing/investment/assets
Renewable Energy	Clean/ Green/ Renewable energy
Climate-Related Topics	climate change, global warming, extreme weather

(Table 1)

4.2.2 ESG Premium

The ESG Premium (*ESGP*) is defined to be the cumulative excess return of ESG oriented company relative to the broader market, using the difference between the return of Amundi PEA MSCI USA ESG Leaders UCITS ETF and the S&P 500. The ESG Leaders ETF exclusively selects U.S.-listed companies with the top 5% ESG ratings based on MSCI ESG scores, which is widely recognized as a leading indicator of ESG investment performance.

We first calculate ESG excess return as follows:

$$ESG\ excess\ return = \left(R_{ESG} - R_f\right) - \beta_{ESG} * \left(R_m - R_f\right) \quad \text{(Eq.1)}$$

Given that this study employs a Vector Error Correction Model (VECM) to analyze the long-term dynamic relationships between variables, ESG excess return is transformed into an asset price representation using continuous compounding:

$$ESG\ Premium_t = \left(\prod_{t=1}^{T}(1 + ESG\ excess\ return_t)\right) - 1 \quad \text{(Eq.2)}$$

4.3 Stationarity and Cointegration Analysis

To ensure the validity of time-series estimation, stationarity is first examined using the Augmented Dickey-Fuller (ADF) test and the Phillips-Perron (PP) test to determine whether ESG search interest and $ESG\ Premium_t$ contain unit roots. If both tests show the variables are stationary at levels, a Vector Autoregression (VAR) model is employed. However, if they are non-stationary, the Johansen cointegration test is conducted to assess the presence of a long-run equilibrium relationship.

4.4 Vector Error Correction Model (VECM)

When a cointegration relationship is confirmed, the VECM is employed to model both short-term fluctuations and long-term equilibrium dynamics.

$$\Delta X_t = \Pi X_{(t-1)} + \sum\left(\Gamma_i \Delta X_{(t-i)}\right) + \varepsilon_t \quad \text{(Eq. 3)}$$

$$\Delta X_t = \Pi X_{t-1} + \sum \Gamma_i\, \Delta X_{t-i} + \Phi \Delta VIX_t + \varepsilon_t \quad \text{(Eq. 4)}$$

In the above equation (Eq.3), X_t represents the vector of endogenous variables, including ESG search trends and $ESG\ Premium_t$. The term Π serve as the error correction component, reflecting the speed at which deviations from equilibrium are adjusted over time, while Γ_i captures short-term effects. The stochastic error term ε_t, accounts for random disturbances in the system. To account for market uncertainty,

Eq.4 is the version that incorporates the VIX index as a control variable.
The test is conducted for multiple lag lengths, with the optimal lag selection determined by the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC).

4.5 Robustness Check: Granger Causality Test

To further examine the robustness of the results of VECM-VAR model, the Granger causality test is conducted. This method evaluates whether past values of Google search trends contain statistically significant information for predicting ESG excess returns. The test follows the structure:

$$Y_t = \alpha + \sum \beta_i Y_{t-i} + \sum \gamma_j X_{t-j} + \sum \delta_k VIX_{t-k} + \varepsilon_t \qquad \text{(Eq.5)}$$

where Y_t represents ESG excess return, X_t denotes ESG search interest and VIX_t is included as a control variabl. The null hypothesis states that past values of ESG-related search trends do not contribute to the prediction of ESG excess returns. If the null hypothesis is rejected, it provides evidence that online search behavior may serve as a leading indicator of ESG investment demand.

5. Results and Discussion

5.1 Stationarity Tests and Model Selection

The initial research plan hypothesized that several of the time series variables, particularly the ESG Premium (gp) and three of the attention indices, would be non-stationary, thus requiring the use of a Vector Error Correction Model (VECM) to analyze their long-run cointegrating relationship .

However, the empirical results from the full dataset (January 2020 – January 2025) contradict this initial assumption. As shown in **Table 2**, the Augmented Dickey-Fuller (ADF) tests for all variables—the four ESG attention indices, the ESG Premium (gp), and the VIX control—yielded p-values significantly below 0.01. Furthermore, the KPSS tests for all variables returned p-values greater than 0.05 (0.1 in all cases), failing to reject the null hypothesis of stationarity.

This is a critical finding: The data confirms that all time series are **stationary at levels (I(0))**.

This result implies that while the variables fluctuate, they consistently revert to a constant mean and do not possess a stochastic trend. The absence of unit roots across all variables means that a cointegration analysis is not appropriate. Therefore, the methodological approach was pivoted from the proposed VECM to a **Vector Autoregression (VAR) model**. This adjustment aligns with the research plan's contingency to use VAR for stationary variables to examine short-term interactions. All subsequent analyses are based on this VAR framework, with the optimal lag length for each model determined by the Akaike Information Criterion (AIC).

Variable	**ADF p-value**	**KPSS p-value**	**Conclusion**
gp (ESG Premium)	6.208e-19	0.1	Stationary (I(0))
general_esg_interest	7.821e-27	0.1	Stationary (I(0))
esg_investment	4.393e-12	0.1	Stationary (I(0))
renewable_energy	1.511e-09	0.1	Stationary (I(0))
climate_topics	3.953e-27	0.1	Stationary (I(0))
vix	3.943e-17	0.1	Stationary (I(0))

(**Table 2**)

5.2 Granger Causality Analysis

To investigate the predictive relationship between public attention and the ESG Premium, a series of pairwise Granger causality tests were conducted. Each test included the VIX index as a control variable to isolate the effects of market uncertainty. The results, summarized in **Table 3**, reveal a clear and distinct pattern.

The analysis provides a direct answer to the research question: "How does the impact vary across different ESG subtopics?". The findings show that **only the investment-oriented attention topics** hold predictive power:

- **Significant Causality:** Both general_esg_interest ($p = 0.047$) and esg_investment ($p = 0.017$) were found to Granger-cause the ESG Premium (gp). This suggests that search activity for terms directly related to ESG as a financial product or concept serves as a leading indicator for short-term excess returns.
- **No Causality:** Conversely, the broader environmental topics, renewable_energy ($p = 0.715$) and climate_topics ($p = 0.462$), showed no statistically significant ability to predict gp.

Furthermore, the tests for reverse causality (i.e., whether gp Granger-causes attention) were **not significant for any of the four topics** (**Table 4**). This finding refutes the hypothesis that rising ESG stock prices create a feedback loop that drives increased investor interest. The relationship appears to be a **unidirectional path from attention to returns**, not a reciprocal one.

Test 1: Attention Granger-cause Premium

Null Hypothesis (H_0)	F-statistic	p-value	Significant ($\alpha = 0.05$)
general_esg_interest ↛ gp	2.6596	0.0472	Yes
esg_investment ↛ gp	4.1165	0.0167	Yes
renewable_energy ↛ gp	0.4536	0.7148	No
climate_topics ↛ gp	0.7729	0.462	No

(Table 3)

Test 2: Premium does not Granger-cause Attention

Null Hypothesis (H_0)	F-statistic	p-value	Significant ($\alpha = 0.05$)
gp ↛ general_esg_interest	0.9238	0.4288	No
gp ↛ esg_investment	0.3756	0.687	No
gp ↛ renewable_energy	1.1952	0.3106	No
gp ↛ climate_topics	1.104	0.3321	No

(Table 4)

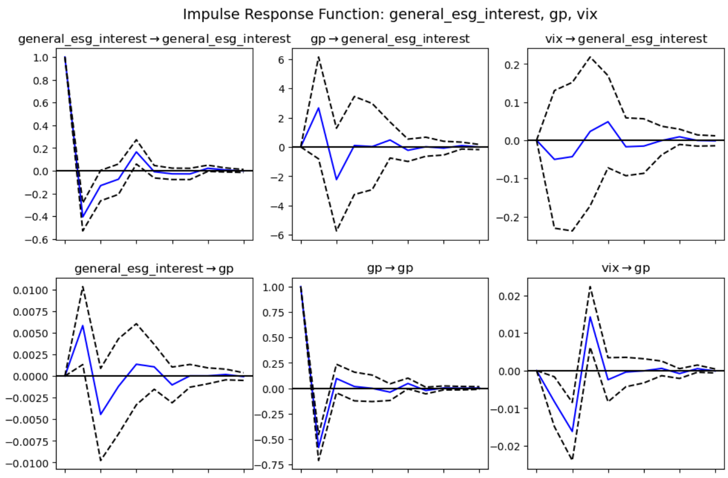
5.3 Impulse Response Function (IRF) Analysis

To visualize the dynamic impact of these causal relationships, Impulse Response Functions (IRFs) were generated from the VAR models. These functions map the response of the ESG Premium (gp) to a one-standard-deviation shock in each attention index over a 10-week horizon.

The IRF plots (**Figures 2-5**) strongly reinforce the findings from the Granger causality tests.

- **Investment-Oriented Topics (Figure 2 & 3):** As seen in the general_esg_interest to gp plot, a shock to "General ESG Interest" causes an immediate, statistically significant positive increase in the ESG Premium at lag 1. The confidence interval (dashed lines) is entirely above zero, confirming the significance of this impact. The response quickly dissipates and reverts to the mean, consistent with the attention-driven excess returns hypothesis. The *esg_investment* to *gp* plot shows an identical and equally significant positive response.

- **Environmental Topics (Figure 4 & 5):** In stark contrast, the *renewable_energy* to *gp* and *climate_topics* to *gp* plots show no significant effect. In both cases, the response (blue line) is centered on zero, and the confidence interval (dashed lines) consistently overlaps the zero line at all lags. This indicates that a surge in public concern about climate change or renewable energy does not translate into market-moving demand for ESG assets.



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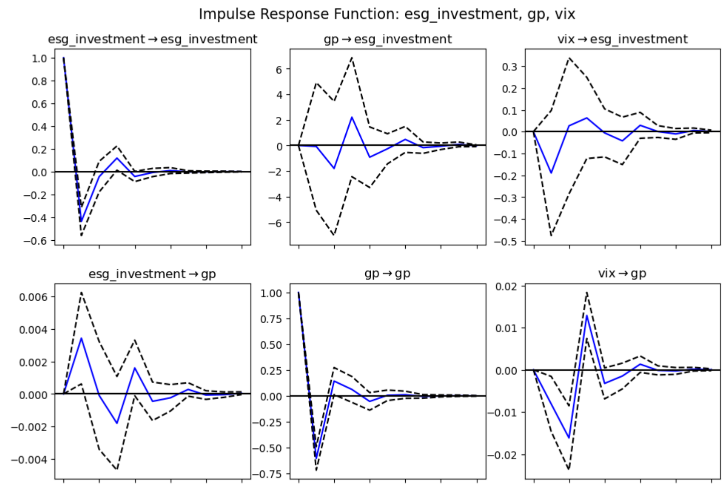

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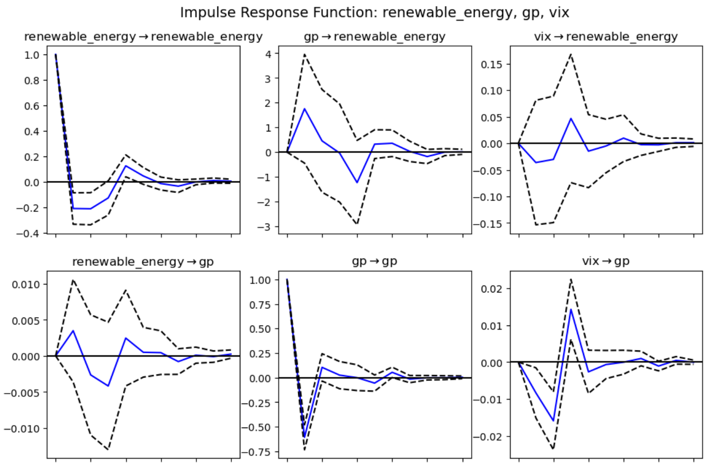

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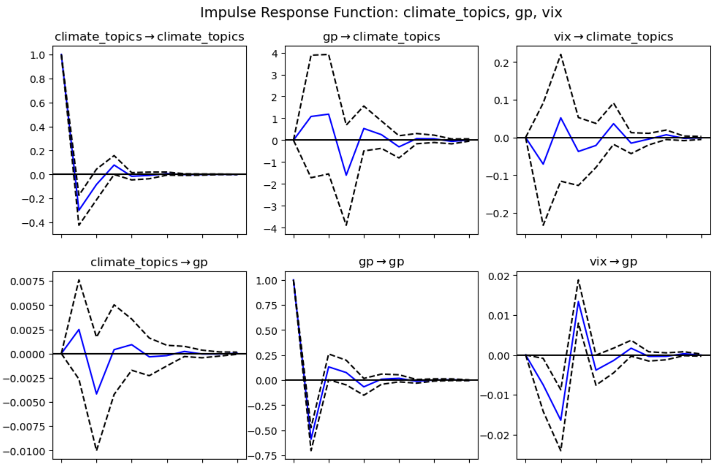

(Figure 5)

Discussion: This "Attention Segmentation" is the central finding of this study. The market effectively distinguishes between public attention as *general environmental concern* (which it ignores) and public attention as *specific investment interest* (which it responds to). This suggests that ESG excess returns are not driven by broad public sentiment about climate change, but rather by the narrower, finance-focused segment of the public actively searching for ESG investment vehicles.

6. Conclusion

6.1 Summary of Findings

This study investigated the dynamic relationship between public attention to ESG subtopics and the excess returns of high-ESG-rated firms. The research aimed to determine if this relationship exhibits short-term dynamics and whether the impact varies by topic.

The empirical analysis yielded four key findings:

1. **Model Specification:** Contrary to initial expectations of non-stationarity, all variables in the study were found to be stationary at levels. This finding led to the rejection of the VECM framework, as no long-term equilibrium relationship exists. The appropriate model for this analysis was a VAR, focusing purely on short-term dynamics.
2. **An " Segmentation in Attention ":** The impact of public attention on ESG returns varies significantly by subtopic. A statistically significant, positive relationship was found for investment-oriented topics (general_esg_interest and esg_investment). However, broader environmental topics (renewable_energy and climate_topics) were found to have no significant impact on excess returns.
3. **Unidirectional Causality:** The relationship is strictly one-way. Investment-specific attention Granger-causes ESG returns. However, the reverse was not true; ESG returns do not Granger-cause public attention in any category. This refutes the hypothesis of a self-reinforcing feedback loop.
4. **Short-Term Impact:** As new attention-driven capital flows into high-ESG-rated firms, it generates positive short-term excess returns, consistent with the findings of El Ouadghiri et al. (2019). The IRFs show this effect is immediate (peaking at lag 1) and transient, dissipating within a few weeks.

6.2 Implications

The findings offer valuable insights for multiple stakeholders.

- **For Investors and Fund Managers:** The results provide a practical, data-driven tool. Google Trends data for *investment-specific* keywords (e.g., "ESG investing") can serve as a potent leading indicator for short-term price movements. Conversely, managers can confidently treat surges in general climate-related news as market "noise" rather than a tradable signal.
- **For Academic Research:** This study contributes to the literature by moving

beyond a monolithic view of "ESG attention." It demonstrates that the *type* of attention is paramount and that segmenting attention indices is crucial for model accuracy. It refines the attention hypothesis in the ESG context, showing that the "attention-to-demand" channel is only active when search behavior is explicitly financial.

6.3 Limitations and Future Research

This study is subject to several limitations. First, Google Trends data is a proxy for *attention*, not necessarily *sentiment* or *intent* . A surge in "ESG investing" searches could be driven by both positive (opportunity) and negative (greenwashing fears) catalysts. Future research could integrate Natural Language Processing (NLP) of social media or news data to dissect the sentiment behind the search volume.

Second, the study period (2020–2025) was characterized by high volatility, including the COVID-19 pandemic and subsequent inflationary period, which may have amplified ESG demand. Repeating the analysis in a different market regime could test the stability of these findings.

Finally, while this study confirms a short-term positive return, it did not find a significant *negative reversal* (market overreaction) as hypothesized. The IRFs simply show a return to the mean. Future research using daily data might provide a higher-resolution view of this mechanism, clarifying whether the effect is simple dissipation or a true, albeit brief, overreaction.

7. References

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Appendix A – Full VAR Regression Tables (Controlling for VIX)

Model 1: General ESG Interest vs GP (controlling for VIX)

N=256 Lag=3 AIC=-15.727 HQIC=-15.560 BIC=-15.312

- Dependent variable: general_esg_interest

Variable	Coefficient	Std. Error	t-stat	p-value
const	0.004874	0.015546	0.314	0.754

L1.general_esg_interest	-0.406353	0.062696	-6.481	0.000
L1.gp	2.659552	1.773919	1.499	0.134
L1.vix	-0.050187	0.092093	-0.545	0.586
L2.general_esg_interest	-0.312684	0.065198	-4.796	0.000
L2.gp	0.374951	1.907080	0.197	0.844
L2.vix	-0.044783	0.092622	-0.484	0.629
L3.general_esg_interest	-0.245955	0.062989	-3.905	0.000
L3.gp	-0.051889	1.648907	-0.031	0.975
L3.vix	0.031360	0.099065	0.317	0.752

- Dependent variable: gp

Variable	Coefficient	Std. Error	t-stat	p-value
const	0.000893	0.000571	1.565	0.118
L1.general_esg_interest	0.005841	0.002302	2.537	0.011
L1.gp	-0.583305	0.065145	-8.954	0.000
L1.vix	-0.008290	0.003382	-2.451	0.014
L2.general_esg_interest	0.001124	0.002394	0.470	0.639
L2.gp	-0.264779	0.070035	-3.781	0.000
L2.vix	-0.021296	0.003401	-6.261	0.000
L3.general_esg_interest	-0.001515	0.002313	-0.655	0.512
L3.gp	-0.085394	0.060554	-1.410	0.158
L3.vix	0.001210	0.003638	0.333	0.739

- Dependent variable: vix

Variable	Coefficient	Std. Error	t-stat	p-value
const	0.015360	0.010910	1.408	0.159
L1.general_esg_interest	-0.025195	0.043997	-0.573	0.567
L1.gp	-0.538385	1.244866	-0.432	0.665
L1.vix	-0.068081	0.064627	-1.053	0.292
L2.general_esg_interest	-0.003479	0.045753	-0.076	0.939
L2.gp	-0.570819	1.338313	-0.427	0.670
L2.vix	-0.044777	0.064998	-0.689	0.491
L3.general_esg_interest	-0.016372	0.044203	-0.370	0.711
L3.gp	1.418412	1.157137	1.226	0.220
L3.vix	-0.029448	0.069520	-0.424	0.672

Model 2: ESG Investment vs GP (controlling for VIX)

N=257 Lag=2 AIC=-14.803 HQIC=-14.686 BIC=-14.513

- Dependent variable: esg_investment

Variable	Coefficient	Std. Error	t-stat	p-value
const	-0.001150	0.024644	-0.047	0.963
L1.esg_investment	-0.436284	0.063011	-6.924	0.000
L1.gp	-0.075651	2.546222	-0.030	0.976
L1.vix	-0.189714	0.146094	-1.299	0.194
L2.esg_investment	-0.237266	0.063488	-3.737	0.000
L2.gp	-1.962939	2.527353	-0.777	0.437
L2.vix	-0.070112	0.146368	-0.479	0.632

- Dependent variable: gp

Variable	Coefficient	Std. Error	t-stat	p-value
const	0.000933	0.000562	1.661	0.097
L1.esg_investment	0.003430	0.001436	2.388	0.017
L1.gp	-0.604720	0.058029	-10.421	0.000
L1.vix	-0.008101	0.003329	-2.433	0.015
L2.esg_investment	0.003323	0.001447	2.297	0.022
L2.gp	-0.226210	0.057599	-3.927	0.000
L2.vix	-0.020989	0.003336	-6.292	0.000

- Dependent variable: vix

Variable	Coefficient	Std. Error	t-stat	p-value
const	0.014970	0.010732	1.395	0.163
L1.esg_investment	-0.019107	0.027441	-0.696	0.486
L1.gp	-0.461204	1.108876	-0.416	0.677
L1.vix	-0.070723	0.063624	-1.112	0.266
L2.esg_investment	-0.021295	0.027649	-0.770	0.441
L2.gp	-1.365415	1.100659	-1.241	0.215
L2.vix	-0.051351	0.063743	-0.806	0.420

Model 3: Renewable Energy vs GP (controlling for VIX)

N=256 Lag=3 AIC=-16.570 HQIC=-16.403 BIC=-16.155

- Dependent variable: renewable_energy

Variable	Coefficient	Std. Error	t-stat	p-value
const	-0.001912	0.010020	-0.191	0.849
L1.renewable_energy	-0.208821	0.062944	-3.318	0.001
L1.gp	1.754538	1.120871	1.565	0.118
L1.vix	-0.036271	0.059727	-0.607	0.544
L2.renewable_energy	-0.256808	0.063255	-4.060	0.000
L2.gp	1.873582	1.223257	1.532	0.126
L2.vix	-0.026053	0.059999	-0.434	0.664
L3.renewable_energy	-0.222671	0.065303	-3.410	0.001
L3.gp	1.445192	1.056488	1.368	0.171
L3.vix	0.071191	0.063774	1.116	0.264

- Dependent variable: gp

Variable	Coefficient	Std. Error	t-stat	p-value
const	0.000931	0.000579	1.609	0.108
L1.renewable_energy	0.003495	0.003634	0.962	0.336
L1.gp	-0.607461	0.064715	-9.387	0.000
L1.vix	-0.008295	0.003448	-2.405	0.016
L2.renewable_energy	0.001029	0.003652	0.282	0.778
L2.gp	-0.273867	0.070627	-3.878	0.000
L2.vix	-0.021397	0.003464	-6.177	0.000
L3.renewable_energy	-0.001717	0.003770	-0.455	0.649
L3.gp	-0.092011	0.060998	-1.508	0.131

L3.vix	0.000607	0.003682	0.165	0.869

- Dependent variable: vix

Variable	Coefficient	Std. Error	t-stat	p-value
const	0.015387	0.010869	1.416	0.157
L1.renewable_energy	0.098372	0.068277	1.441	0.150
L1.gp	-0.391480	1.215843	-0.322	0.747
L1.vix	-0.075308	0.064788	-1.162	0.245
L2.renewable_energy	0.035386	0.068615	0.516	0.606
L2.gp	-0.760049	1.326905	-0.573	0.567
L2.vix	-0.038665	0.065083	-0.594	0.552
L3.renewable_energy	0.046136	0.070836	0.651	0.515
L3.gp	1.425561	1.146005	1.244	0.214
L3.vix	-0.020976	0.069178	-0.303	0.762

Model 4: Climate Topics vs GP (controlling for VIX)

N=257 Lag=2 AIC=-15.940 HQIC=-15.824 BIC=-15.650

- Dependent variable: climate_topics

Variable	Coefficient	Std. Error	t-stat	p-value
const	-0.001386	0.013857	-0.100	0.920
L1.climate_topics	-0.298714	0.063689	-4.690	0.000
L1.gp	1.084301	1.428256	0.759	0.448
L1.vix	-0.070563	0.082576	-0.855	0.393

L2.climate_topics	-0.170191	0.064157	-2.653	0.008
L2.gp	2.111429	1.422007	1.485	0.138
L2.vix	0.034020	0.082259	0.414	0.679

- Dependent variable: gp

Variable	Coefficient	Std. Error	t-stat	p-value
const	0.000886	0.000569	1.558	0.119
L1.climate_topics	0.002483	0.002615	0.950	0.342
L1.gp	-0.590006	0.058649	-10.060	0.000
L1.vix	-0.007516	0.003391	-2.217	0.027
L2.climate_topics	-0.001392	0.002635	-0.528	0.597
L2.gp	-0.223457	0.058392	-3.827	0.000
L2.vix	-0.021189	0.003378	-6.273	0.000

- Dependent variable: vix

Variable	Coefficient	Std. Error	t-stat	p-value
const	0.015438	0.010693	1.444	0.149
L1.climate_topics	0.076356	0.049146	1.554	0.120
L1.gp	-0.598276	1.102123	-0.543	0.587
L1.vix	-0.072711	0.063720	-1.141	0.254
L2.climate_topics	0.042451	0.049507	0.857	0.391
L2.gp	-1.390535	1.097301	-1.267	0.205
L2.vix	-0.041448	0.063476	-0.653	0.514